Roll No. ....

TBT–501

B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

RECOMBINANT DNA TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What do you understand from Biosafety ? Describe layout of BSL2 facility with importance features. (CO1)

OR

(b) According to you, what are 10 landmark discoveries in RDT ? (CO1)

2. (a) Discuss scope and application of RDT. (CO1) & (CO2)

OR

(b) Why Type-II restriction endonucleases are perferred in RDT ? (CO1) & (CO2)

3. (a) Write short notes on the following : (CO2)

(i) S–1 endonuclease

(ii) Linkers

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OR

(b) What are modifying enzymes ? Write application of any *one* in gene cloning. (CO2)

4. (a) How would you generate two different sticky ends onto any insert DNA ? (CO2) & (CO3)

OR

(b) What is vector ? How are positive clones screened in pBR322 ?

(CO2) & (CO3)

5. (a) Draw vector pUC18 and describe cloning and screening strategies in it.

(CO2) & (CO3)

OR

(b) Write short notes on the following : (CO2) & (CO3)

(i) Blue white screening

(ii) α-Complementation

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TME–501 (A)

B. TECH (BIOTECHNOLOGY) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

HEAT AND MASS TRANSFER

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are the mechanisms of heat transfer ? How are they distinguished from each other ? (CO1)

OR

(b) A furnace wall is composed of 220 mm of fire brick, 150 mm of common brick, 50 mm of 80 % magnesia and 3 mm of steel plate on the outside. If the inside surface temperature is 1500°C and outside surface temperature is 90°C, estimate the temperature between layers and calculate the heat loss. Assume, K (for fire brick) = 1.31 W/m-°C, K (for common brick) = 0.61 W/m-°C, K (for 85% magnesia) = 0.71 W/m-°C, K (steel) = 66.67 W/m-°C. (CO1)

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2. (a) Derive 3-dimensional heat transfer equation in Cartesian coordinate system. (CO1)

OR

(b) A steam pipe of 5 cm inside diameter and 6.5 cm outside diameter is covered with a 2.75 cm radial thickness of high temperature insulation (K = 1.1 W/m-K). The surface heat transfer coefficients for outside and inside surfaces are 11.5 W/m^2-K and 4650 W/m^2-K. The thermal conductivity of pipe material is 45 W/m-K If the steam temperature is 200°C and ambient temperature is 25°C, determine heat transfer per m length of pipe, temperature at the interface and overall heat transfer coefficient. (CO1)

3. (a) What is the significance of heat transfer ? What do you mean by thermal conductivity ? Why good electrical conductor materials are also good heat conductors ? Explain with an example. (CO1)

OR

(b) Discuss electrical analogy of combined heat conduction and convection in a composite plane wall. (CO1)

4. (a) What are the reasons for using fins on the surface ? Define effectiveness and efficiency of fin. (CO2)

OR

(b) A stainless-steel fin (K = 20 W/m-K) having diameter of 20 mm and a length of 0.1 m is attached to a wall at 300°C. The ambient temperature is 50°C and heat transfer coefficient is 10 W/m^2-K. The fin tip is insulated. Calculate the rate of heat dissipation from the fin and the temperature at the fin tip. (CO2)

5. (a) State and explain the various laws of heat transfer. Write a short note on the practical application of fin. (CO2)

OR

(b) A hot surface at 100°C is to be cooled by attaching a 3 cm long, 0.25 cm diameter aluminum pin fin (K=237 W/m-°C) to it, with a center-to-center distance of 0.6 cm The temperature of surrounding medium is 30°C, and the heat transfer coefficient on the surface is 35 W/m^2-°C. Determine the rate of heat transfer from the surface for a 1 m × 1 m section of the plate. Also, determine the overall effectiveness of fins. (CO2)

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TBT–502

B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

BIOPROCESS ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Describe the History of the Fermentation Industry in detail. (CO1)

OR

(b) Explain media optimization using Plackett Burman Design. (CO1)

2. (a) What are the attributes of media ? Explain effect of dissolved oxygen and pH on microbial growth kinetics. (CO1)

OR

(b) Discuss thermal death kinetics. Explain X_{90} and its significance. (CO1)

3. (a) Psedomonas aeruginosa has a mass doubling time of 3.6 h when grown on nutrient agar. The saturation constant using this substrate is 2.4 g/l (which is unusually high) and cell yield on nutrient agar is 0.46 g cell/g nutrient agar. If we operate a chemostat on a feed stream containing

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38 g/l nutrient agar and kinetic parameter value of μ_m is 0.42 h^{-1}, find the following :

(i) Cell concentration

(ii) Optimum dilution rate

(iii) Cell productivity when Dilution rate is one fourth of the maximum. (CO2)

OR

(b) Derive an expression for operating condition and biomass concentration in continuous culture using mass balance equation. (CO2)

4. (a) Derive an expression for total biomass concentration in Fed Batch culture. (CO2)

OR

(b) Discuss Monod equation and its applications. What is limiting substrate and cryptic growth. (CO2)

5. (a) Derive an expression for total product concentration in Fed batch culture. (CO2)

OR

(b) What are different types of products obtained in Batch culture ? Explain Q_p in each case. (CO2)

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TBT–503

B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

BIOINFORMATICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What do you understand by Bioinformatics ? Why is it important for a Biotechnologists ? (CO1)

OR

(b) Throw light on NCBI as a free resource for Bioinformatics. (CO5)

2. (a) Elaborate various protocols used in internet. (CO1)

OR

(b) Explain in brief about the following : (CO1, CO4)

(i) Fasta format

(ii) Boolean operators

3. (a) Give contribution of any *four* scientists in Bioinformatics. (CO1, CO5)

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OR

(b) Describe any *two* case studies as contribution of Bioinformatics in Biotechnology. (CO1, CO5)

4. (a) Elaborate various annotations provided in UNIPROT database available for a protein entry. (CO2)

OR

(b) Discuss Restriction Digestion as a laboratory technique for generating data for making Bioinformatics tools and servers. (CO5)

5. (a) Throw light on the following : (CO1)

(i) INSDC

(ii) Model Organism Database

OR

(b) What are biological databases ? Elaborate primary and secondary databases along with examples. (CO2)

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TBT–504(a)

B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

MARINE BIOTECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define Marine Biotechnology and discuss its various scope. (CO1)

OR

(b) Elaborate on the different applications of Marine Biotechnology. (CO1)

2. (a) Discuss some of the economically important oysters in India. (CO1)

OR

(b) Elaborate on some of the economically important fin fishes in India. (CO1)

3. (a) Write a note on the varied microbial diversity in a marine ecosystem. (CO1/ CO2)

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OR

(b) What are extremophiles ? Discuss in detail the different kinds of extremophiles. (CO1/ CO2)

4. (a) What is a non-culturable microbe ? Discuss, how can metagenomics be studied to identify such microbes. (CO2)

OR

(b) Discuss the protocol for collection, isolation, screening and identification in marine samples. (CO2)

5. (a) Elaborate on the different roles microbes play in a ecosystem. (CO2)

OR

(b) Discuss the protocol for collection isolation, screening, and identification of fungi in marine samples. (CO2)

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TBT–504(b)

B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

STEM CELL BIOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Review the milestones of advances in stem cell biology. (CO1)

OR

(b) Describe neural stem cells and its role in neurogenesis. (CO1)

2. (a) Explain induced pluripotent stem cell. (CO1, CO2)

OR

(b) Explain the different methods for fate mapping of stem cells. (CO1, CO2)

3. (a) Discuss adult stem cells and its applications. (CO1, CO2)

OR

(b) Review mesenchymal stem cells and its differentiation potential. (CO1, CO2)

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4. (a) Elaborate on cancer stem cells and the different signaling pathways. (CO2)

OR

(b) Describe embryonic stem cell with emphasis on its production and maintenance. (CO2)

5. (a) Define stem cells and discuss the general properties of stem cells. (CO2)

OR

(b) Briefly describe the tests used for the identification of stem cells. (CO2)

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TBT–505B

B. TECH. (BIOTECH.) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

3D PRINTING AND DESIGNING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) How has 3D printing revolutionized the manufacturing industry, and what are some notable examples of products or components that have been successfully produced using 3D printing ? (CO1)

OR

(b) Explain how additive manufacturing changes current scenario of manufacturing world. (CO1)

2. (a) Classify additive manufacturing systems. Explain each system in detail. (CO1)

OR

(b) Can you explain the advantages and limitations of SLS technology compared to other 3D printing methods, such as Fused Deposition Modeling (FDM) or Stereolithography (SLA) ? (CO1)

3. (a) How has FDM 3D printing technology evolved over the years, and what are some recent advancements or applications that highlight its versatility and potential impact in various industries ? (CO1)

OR

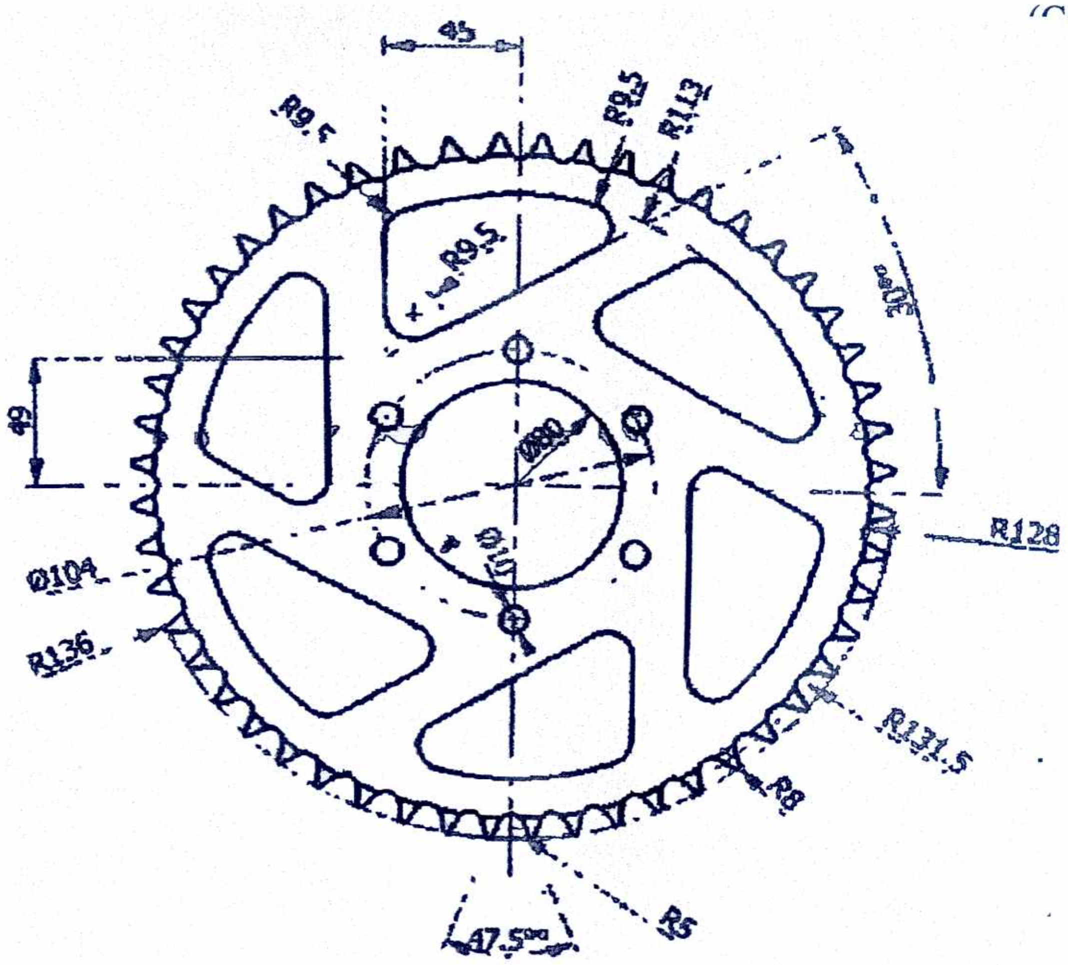
(b) How does the SLS 3D printing process vary from traditional manufacturing methods in terms of speed, cost-effectiveness, and design flexibility, and what are some challenges that users might encounter when working with SLS technology ? (CO1)

4. (a) Can you explain the common methodologies and tools used in reverse engineering software applications, including both static and dynamic analysis techniques ? (CO2)

OR

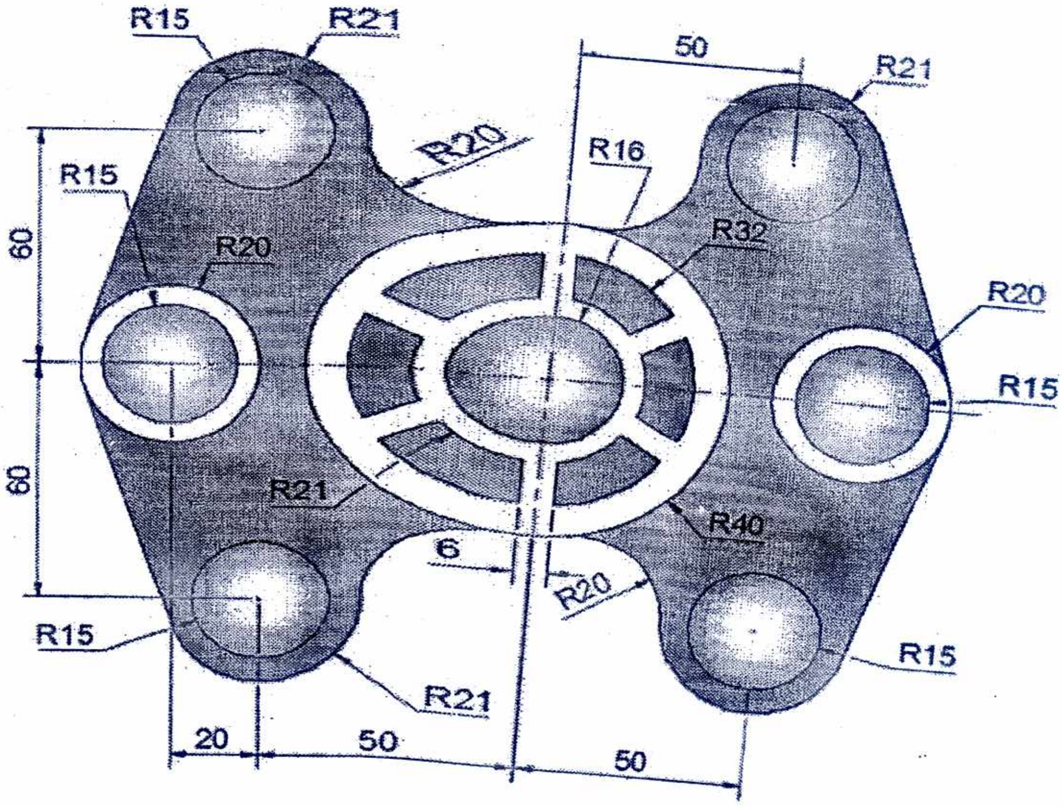
(b) In which fields you see the maximum use of CAD software ? Explain with suitable example. (CO2)

5. (a) Write the steps involve in making given drawing on AutoCad software. (CO2)



OR

(b) Write the steps involved in making given drawing on AutoCad Software : (CO2)



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TBT–506(b)

B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

NETWORK BIOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What do you understand by Systems Biology ? Elaborate diagrammatically the interactions between various levels of biological organizations. (CO2)

OR

(b) What do you understand by computer networks and network topology ? Differentiate between physical and logical topology. (CO1)

2. (a) What do you understand by graph theory ? What are graphs ? Also throw light on various applications of Graph Theory. (CO1)

OR

(b) What are connectivity maps ? What are its applications ? (CO2)

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3. (a) Throw light on various types of networks in biology along with proper diagram : (CO1)

    (i) Protein-protein network

    (ii) Protein-DNA network

    (iii) Metabolic networks

OR

   (b) Elaborate the concept of lethality and centrality in protein networks along with proper example. (CO1)

4. (a) Define protein-protein interactions. Also throw light on applications of protein-protein interaction studies. (CO1)

OR

   (b) What do you understand by specificity in protein interaction networks ? (CO2)

5. (a) Describe STRING database for protein-protein interaction networks. (CO2)

OR

   (b) Give any *five* examples of protein-protein interaction in humans along with proper diagram. (CO2)